Two Analysts, One Planned Test

Topic 6.10 · TODAY'S PROBLEM

Suppose an online retailer has 50,000 active accounts. Before collecting data, its team asks, "Does a progress meter change the proportion of active accounts completing checkout?" It randomly selects 360 accounts and assigns 180 to each screen. The table shows the outcomes. Let p_M and p_S be the corresponding population completion proportions.

Checkout outcomes

Screen	n	Complete	Other
Meter	180	126	54
Standard	180	108	72

Nia: "Use $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$ because either direction is a change."

Omar: "Because $\hat{p}_M$ is larger, use $H_a : p_M - p_S > 0$."

FIRST TAKE (5 min, on your sheet)

Which analyst's setup should the team use? Cite one phrase or timing detail from the study.

DOK 1 **(a)** Define p_M and p_S in context. Calculate $\hat{p}_M$, $\hat{p}_S$, and the observed difference $\hat{p}_M - \hat{p}_S$.

DOK 2 **(b)** Name the test procedure. Calculate the pooled proportion and verify randomization, the 10 percent condition, and all four pooled expected counts.

DOK 3 **(c)** Decide which setup is defensible and state the hypotheses. Use the word 'change,' the timing of the tail choice, condition evidence, and both kinds of randomization. Explain how the rejected setup would misrepresent the strength or scope of the evidence.

Video + follow-along: `u6_lesson10_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.